Module 10-1 to 10-4 #1

Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **45**

Instructor Name: **Bodanske, Daniel**

Question 1 of 45

Evaluate the expressions.

$-(9)^0 =$ _____

$2\left(-\frac{3}{7}\right)^0 =$ _____

Question 2 of 45

Simplify.

$\sqrt{32}$

Question 3 of 45

Simplify.

$\sqrt{128}$

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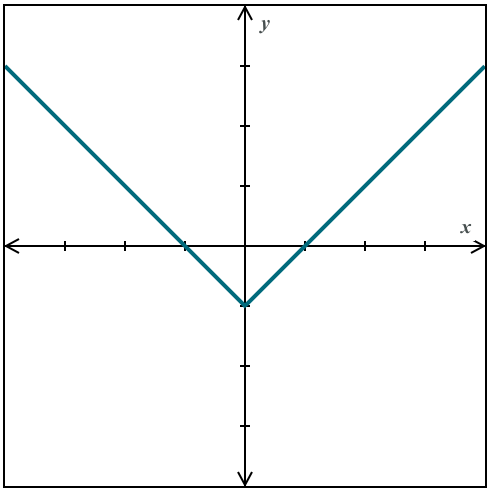
Question 4 of 45

Four functions are given below. Either the function is defined explicitly, or the entire graph of the function is shown.

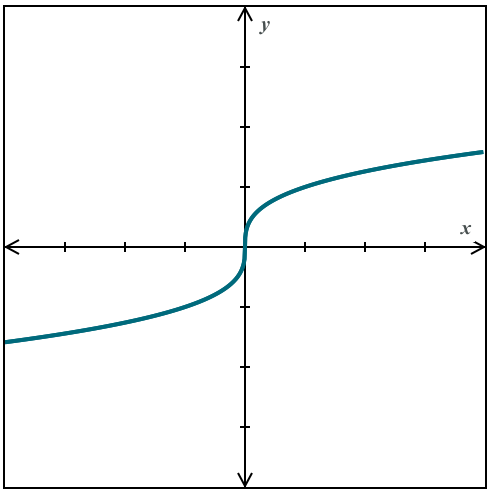
For each, decide whether it is an even function, an odd function, or neither.

The function r



- Even - Odd - Neither

The function s



- Even - Odd - Neither

$g(x) = 7x^4 - 2x^3$

- Even - Odd - Neither

$h(x) = 7x^2$

- Even - Odd - Neither

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Question 5 of 45

Multiply.

$$-v^5\left(-4v^5\right)$$

Simplify your answer as much as possible.

Question 6 of 45

Multiply.

$$4y^2\cdot 4v\cdot 3v^7y^7$$

Simplify your answer as much as possible.

Question 7 of 45

Simplify.

$$\left(w^3\right)^3$$

Write your answer without parentheses.

Question 8 of 45

Simplify.

$$\left(3u^2w^4\right)^4$$

Write your answer without parentheses.

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Question 9 of 45

Simplify.

$$\left(3x^3 y^3 z\right)^2 \left(y^3 z^2\right)$$

Question 10 of 45

Write 6×10^5 in standard notation.

Question 11 of 45

Write 6.363×10^5 in standard notation.

Question 12 of 45

Calculate.

$$\left(3 \times 10^9\right)\left(2 \times 10^6\right)$$

Write your answer in scientific notation.

Question 13 of 45

Light travels 1.08×10^9 kilometers in one hour. How far does it travel in 4 hours?

Write your answer in scientific notation.

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Question 14 of 45

Simplify.

$$\left(\frac{w^3}{9v^4}\right)^2$$

Write your answer without parentheses.

Question 15 of 45

Simplify.

$$\frac{a^5}{a^2}$$

Question 16 of 45

Simplify.

$$\frac{18x^2}{6x^3}$$

Question 17 of 45

Simplify.

$$\frac{z^5 x^6}{z^4 x}$$

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Question 18 of 45

Simplify.

$$\frac{8w^5x^5}{48w^2x}$$

Question 19 of 45

Simplify.

$$\left(\frac{2a^4b^3}{6a^2}\right)^3$$

Write your answer using only positive exponents.

Question 20 of 45

Evaluate the expressions.

$$-3\left(\frac{3}{5}\right)^0 = _____$$

$$(-7)^0 = _____$$

Question 21 of 45

Rewrite the following without an exponent.

$$2^{-4}$$

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Question 22 of 45

Rewrite the following without an exponent.

$$\left(\frac{2}{3}\right)^{-2}$$

Question 23 of 45

Rewrite the following without an exponent.

$$(-3)^{-3}$$

Question 24 of 45

Rewrite the expression without using a negative exponent.

$$3n^{-4}$$

Simplify your answer as much as possible.

Question 25 of 45

Simplify.

$$v^{-4} \cdot v^{5} \cdot v^{-2}$$

Write your answer with a positive exponent only.

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Question 26 of 45

Simplify.

$$\frac{x^{-2}}{x}$$

Write your answer with a positive exponent only.

Question 27 of 45

Simplify.

$$\frac{2n^{-6}m^{2}}{10m^{-8}n^{-5}}$$

Write your answer using only positive exponents.

Question 28 of 45

Simplify.

$$\left(v^{-5}\right)^{-7}$$

Write your answer without using negative exponents.

Question 29 of 45

Simplify.

$$\left(2w^{4}x^{-2}\right)^{-3}$$

Write your answer using only positive exponents.

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Question 30 of 45

Simplify.

$$\left(\frac{4y^{5}}{y^{-2}}\right)^{-3}$$

Write your answer using only positive exponents.

Question 31 of 45

Simplify.

$$\left(\frac{2a^{-4}b^{-1}c^{-8}}{6a^{-6}b^{-2}c}\right)^{-1}$$

Write your answer using only positive exponents.

Question 32 of 45

Write 8×10^{-6} in standard notation.

Question 33 of 45

Write 2.291×10^{-3} in standard notation.

Question 34 of 45

Answer the following.

(a) A certain solution has a hydrogen ion concentration of 3.54×10^{-5} moles per liter. Write this number in standard notation.

(b) A hippopotamus can weigh up to 9900 pounds. Write this number in scientific notation.

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Question 35 of 45

Calculate.

$$\frac{7.7 \times 10^{8}}{1.4 \times 10^{-7}}$$

Write your answer in scientific notation.

Question 36 of 45

Find the value of $\sqrt[3]{1000}$.

Question 37 of 45

Evaluate the following.

(a) $\sqrt[4]{-256}$ = _____

(b) $\sqrt[3]{-27}$ = _____

Question 38 of 45

Simplify.

$$\sqrt[4]{\frac{16}{625}}$$

Be sure to write your answer in lowest terms.

Question 39 of 45

Write the following as an exponential expression.

$$\sqrt[7]{b^{3}}$$

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Question 40 of 45

Complete the parts below to define $6^{\frac{1}{3}}$.

(a) Let's see what happens when we raise $6^{\frac{1}{3}}$ to an exponent of 3. Use the product of powers rule to fill in the blanks.

$$\left(6^{\frac{1}{3}}\right)^3 = 6^{\frac{1}{3}} \cdot 6^{\frac{1}{3}} \cdot 6^{\frac{1}{3}} = 6^{\frac{1}{3}+___+___} = 6^{___} = ___$$

(b) If we want the product of powers rule to hold true, which of these values should we define $6^{\frac{1}{3}}$ to be?

- 6
- 6^2
- 6^3
- $\sqrt{6}$
- $\sqrt[3]{6}$

Question 41 of 45

Evaluate.

$16^{\frac{1}{4}} = ____$

$27^{\frac{1}{3}} = ____$

Evaluate.

$16^{\frac{1}{4}} = _____$

$27^{\frac{1}{3}} = _____$

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Question 42 of 45

Simplify.

$$16^{\frac{3}{2}}$$

Question 43 of 45

Simplify. Write your answers without exponents.

$$81^{-\frac{3}{4}} = _____$$

$$\left(\frac{1}{32}\right)^{\frac{3}{5}} = _____$$

Question 44 of 45

Simplify.

$$y^{\frac{7}{8}} \cdot y^{\frac{3}{4}}$$

Assume that the variable represents a positive real number.

Question 45 of 45

Simplify.

$$\left(w^{7}\right)^{\frac{5}{6}}$$

Write your answer without parentheses.
Assume that the variable represents a positive real number.

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **45**

Question 1 of 45

$$-(9)^0 = -1$$

$$2\left(-\frac{3}{7}\right)^0 = 2$$

Question 2 of 45

$4\sqrt{2}$

Question 3 of 45

$8\sqrt{2}$

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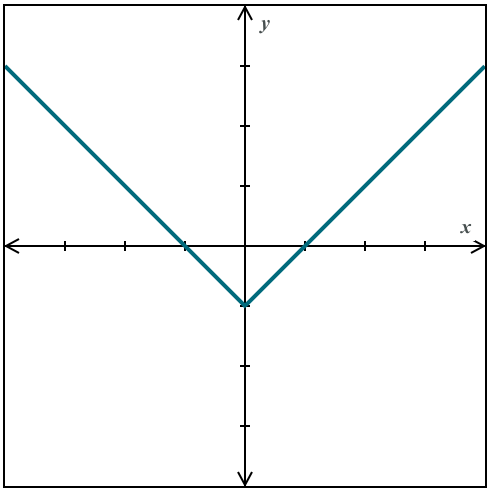
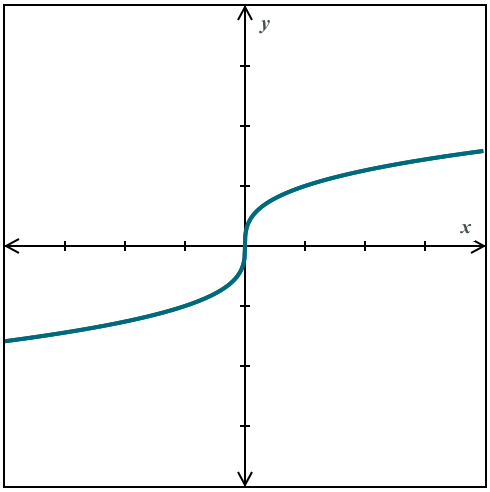
Question 4 of 45

The function r



- Even

The function s



- Odd

$g(x) = 7x^4 - 2x^3$

- Neither

$h(x) = 7x^2$

- Even

Question 5 of 45

$4v^{10}$

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Question 6 of 45

$48y^9 v^8$

Question 7 of 45

w^9

Question 8 of 45

$81u^8 w^{16}$

Question 9 of 45

$9x^6 y^9 z^4$

Question 10 of 45

$600{,}000$

Question 11 of 45

$636{,}300$

Question 12 of 45

6×10^{15}

Question 13 of 45

4.32×10^9 kilometers

Question 14 of 45

$$\frac{w^6}{81v^8}$$

Question 15 of 45

a^3

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Question 16 of 45

$$\frac{3}{x}$$

Question 17 of 45

$$z\,x^5$$

Question 18 of 45

$$\frac{w^3\,x^4}{6}$$

Question 19 of 45

$$\frac{a^6\,b^9}{27}$$

Question 20 of 45

$$-3\left(\frac{3}{5}\right)^0 = -3$$

$$(-7)^0 = 1$$

Question 21 of 45

$$\frac{1}{16}$$

Question 22 of 45

$$\frac{9}{4}$$

Question 23 of 45

$$-\frac{1}{27}$$

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Question 24 of 45

$$\frac{3}{n^4}$$

Question 25 of 45

$$\frac{1}{v}$$

Question 26 of 45

$$\frac{1}{x^3}$$

Question 27 of 45

$$\frac{m^{10}}{5n}$$

Question 28 of 45

$$v^{35}$$

Question 29 of 45

$$\frac{x^6}{8w^{12}}$$

Question 30 of 45

$$\frac{1}{64y^{21}}$$

Question 31 of 45

$$\frac{3c^9}{a^2b}$$

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Question 32 of 45

0.000008

Question 33 of 45

0.002291

Question 34 of 45

(a) 0.0000354 moles per liter

(b) 9.9×10^3 pounds

Question 35 of 45

5.5×10^{15}

Question 36 of 45

10

Question 37 of 45

(a) $\sqrt[4]{-256}$ Not a real number

(b) $\sqrt[3]{-27} = -3$

Question 38 of 45

$\frac{2}{5}$

Question 39 of 45

$b^{\frac{3}{7}}$

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Question 40 of 45

(a) $\left(6^{\frac{1}{3}}\right)^3 = 6^{\frac{1}{3}} \cdot 6^{\frac{1}{3}} \cdot 6^{\frac{1}{3}} = 6^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}} = 6^1 = 6$

(b) ○ 6 ○ 6^2 ○ 6^3 ○ $\sqrt{6}$ ◉ $\sqrt[3]{6}$

Question 41 of 45

$16^{\frac{1}{4}} = 2$

$27^{\frac{1}{3}} = 3$

Question 42 of 45

64

Question 43 of 45

$81^{-\frac{3}{4}} = \frac{1}{27}$

$\left(\frac{1}{32}\right)^{\frac{3}{5}} = \frac{1}{8}$

Question 44 of 45

$y^{\frac{13}{8}}$

Question 45 of 45

$w^{\frac{35}{6}}$

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